

Sriram Sankar

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AWARDS & FUNDING

ASTRO3D PhD Scholarship
2024–27

SALT-SAAO Prize MSc Scholarship
SAAO | 2021

Best MSc Oral Presentation
SAIP annual conference | 2022

FORMAL EDUCATION

UNIVERSITY OF WESTERN
AUSTRALIA (UWA)
PhD Physics | 2024–27
Perth, Australia

UNIVERSITY OF CAPE
TOWN (UCT)
MSc Astronomy | 2021–23
Cape Town, South Africa

MAHATMA GANDHI
UNIVERSITY (MGU)
B.Tech Mechanical Engr. |
2014–18
Kottayam, Kerala, India

TECHNICAL SKILLS

Scientific programming:

PYTHON • BASH • C/C++
SLURM • DOCKER • GIT

Astronomy software:

CASA • CARTA • SoFIA2 • GIZMO
^{3D}BAROLO • CLOUDY • YT • ASTROPY

Open source contributions:

PROCESSMEERKAT, GAUSSPY+,
BAYGAUD-PI, YT_ASTRO_ANALYSIS

HPC experience:

[ilifu](#), [Setonix](#), [OzStar](#), [Acacia](#),
[CANFAR](#), [AWS EC2](#), Hyades, Astro3

RESEARCH SUMMARY

I bridge hydrodynamic simulations and resolved H_I observations to study how galaxies accrete gas at the disc–halo interface. I showed that hot accretion explains extended, warped H_I discs, quantified how much anomalous gas standard extraction misses, and developed a method to separate it from rotating discs. I reduced MeerKAT data on two SKA Regional Centre prototypes, won over 90 hours of telescope time as PI, and supervised six students.

PUBLICATIONS

6 published, 3 to be submitted, 2 in prep | h-index 5 | [ADS library](#)

Selected first-author papers

1. **S. Sankar**, C. Power, B. Catinella, and J. Stern “**Bridging simulations and H_I observations at the disc-CGM interface: forward-modelling anomalous gas in FIRE-2 galaxies**” to be submitted (Oct 2026).
2. **S. Sankar**, J. Stern, C. Power, B. Catinella, D. Fielding, C.-A. Faucher-Giguère, I. Sultan, M. Boylan-Kolchin, and J. Bland-Hawthorn “**Hot accretion onto spiral galaxies: the origin of extended and warped H_I discs**” MNRAS 549, no. 4 (Jul 2026): [stag1059](#).
3. **S. Sankar**, A. Narayanan, B.D. Savage, V. Khaire, B.E. Rosenwasser, J.C. Charlton, and B.P. Wakker “**Physical conditions of five O VI absorption systems towards PG 1522 + 101**” MNRAS 498, no. 4 (Nov 2020): 4864–86. *Featured in C.W. Churchill’s [Quasar Absorption Lines](#) textbook as a first detailed view of O VI absorbers at $z \sim 1$.*

Also: Sardar, **Sankar**+ (to be submitted, student-led); Guimarães Silva+ 2026; Cortese+ 2026; Sameer+ 2022; Pradeep, **Sankar**+ 2020.

PROPOSALS & TELESCOPE TIME

As PI (over 90 hours awarded)

- **The first multiphase study of star-formation-driven outflows below the star-forming main sequence – 15 hours (priority A) – 2024 MeerKAT:** H_I, continuum, and polarisation of a backsplash galaxy with a bipolar outflow.
- **MeerRings – PI & Technical Lead, 55 hours – 2023 MeerKAT** programme mapping H_I in seven collisional ring galaxies. I reduced all data (66 hours, 11 archival) with a modified PROCESSMEERKAT pipeline running on [ilifu](#), under an HPC allocation that I secured.
- **SALTChoirs – 21.4 P1 hours – 2021 SALT:** ionised gas in two groups.

As Co-I

- **MeerINDIAN** (PI: E. de Blok) – 1600 hours, 2026 **MeerKAT** XLP proposal (submitted); **AM 0144–585** (PI: Z. Randriamanakoto) – 11.34 P1 hours, 2024 **SALT**; **MeerChoirs** (PI: M. Mogotsi) – 50 hours, 2022 **MeerKAT**.
- **AusSRC Science Enabling Project** (2026, in preparation) – extending MeerKAT data-reduction support to Australian researchers; **Pawsey Partner Allocation Scheme** (2027 round, in preparation).

LEADERSHIP & SERVICE

- **ICRAR/UWA:** Student Representative; Computational Theory Group host
- **SAAO/UCT:** Green SAAO founder; Extragalactic Discussion Group organiser; Postgraduate Student Representative; outreach volunteer (SAAO, Astronomers for Planet Earth)
- **Undergraduate leadership:** TEDxFISAT organiser; ASME chapter chair; led a department portfolio initiative

RESEARCH EXPERIENCE

INTERNATIONAL CENTRE FOR RADIO ASTRONOMY RESEARCH (ICRAR)

ASTRO3D PhD candidate | Feb 2024–Aug 2027 | Perth, Australia

Supervisors: [Prof. Chris Power](#) & [Prof. Barbara Catinella](#)

- I use HI observations and simulations to investigate the physical nature of anomalous gas.
- In [Sankar+2026b](#), I forward-modelled anomalous HI in six Milky Way-mass FIRE-2 galaxies as synthetic 3D cubes and showed that standard peak-masking extraction recovers only a small fraction of the simulated anomalous gas.

PAWSEY SUPERCOMPUTING CENTRE

Summer intern | Nov 2025–Feb 2026 | Perth, Australia

Supervisors: [Gregory Orange](#) & [Luke Edwards](#)

- I was picked from over 700 applicants to intern at one of two Tier-1 High-Performance Computing facilities in Australia.
- I benchmarked four data-transfer tools between Pawsey's Acacia object store and the Setonix scratch file system, informing user documentation, operational improvements, and service offerings.

TEL AVIV UNIVERSITY (TAU)

Visiting research student | Jul–Sep 2023 | Tel Aviv, Israel

Supervisor: [Prof. Jonathan Stern](#)

- I ran idealised hydrodynamic simulations (on AWS EC2) and analytical calculations showing that misaligned hot CGM inflows can produce neutral hydrogen warps in local galaxies ([Sankar+2026a](#)).

SOUTH AFRICAN ASTRONOMICAL OBSERVATORY (SAAO)

[Prize](#) MSc research student | Mar 2021–Jul 2023 | Cape Town, South Africa

Supervisors: [Dr. Moses Mogotsi](#) & [Prof. Matthew A. Bershad](#)

- For my [dissertation](#), I built a MeerKAT data-reduction and HI analysis pipeline and developed a novel method combining 3D tilted-ring modelling, Gaussian decomposition, and kinematic tagging to separate anomalous gas from rotating discs. I applied both to interacting galaxies in two galaxy groups to isolate signatures of gas exchange.

INDIAN INSTITUTE OF SPACE SCIENCE AND TECHNOLOGY (IIST)

Project student | Feb 2019–Jan 2021 | Trivandrum, Kerala

Supervisor: [Prof. Anand Narayanan](#)

- I used multi-component Voigt profile fitting and ionisation modelling to extract the metallicity–density–temperature structure of circumgalactic absorbers (HST, Keck), contributing to three publications ([Sameer+2022](#), [Sankar+2020](#), [Pradeep](#), [Sankar+2020](#)).

STUDENT SUPERVISION & MENTORSHIP

Main supervisor unless noted; Prof. A. Narayanan co-supervises the IIST students.

- [Suvajit Sardar](#) – MSc, IIST (2024–25) and PhD, IIST (2026–present)
- [Reshoketswe Thobejane](#) – PhD, SAAO/UCT (2025–present); co-supervisor, with Dr. M. Mogotsi as main supervisor
- [Raghav Vyas](#) – MSc, IIST (2026–present)
- [Niladri Biswas](#) – MSc, IIST (2025–26)
- [Maneesha P.V.](#) – MSc, Kerala University (2024)
- [Archie Sreekumar](#) – BSc, Kerala University (2024); now an MSc student at Groningen

SELECTED PRESENTATIONS

- **Hot accretion onto spiral galaxies: the origin of extended and warped HI discs** – [The Many Scales of Galaxy Environments](#), Ascona (Jul 2026); SKA Science Meeting, ICRAR (Apr 2026)
- **Tracing anomalous gas across phases, galaxies, and environments** – [ASTRO3D Science Legacy Meeting](#), Sydney (Jun 2024); Astronomical Society of Australia Hub Day (Jun 2024)
- **Anomalous gas safari: insights from MeerKAT's view of galaxy interactions** – SAAO, Cape Town (Jul 2023); [SKA Pathfinder HI Survey Coordination Committee \(PHISCC\)](#), Cape Town (Mar 2023)
- **Neutral gas kinematics of interacting galaxies in two groups** – [Kapteyn Institute](#), University of Groningen (Sep 2022); [What Matters Around Galaxies \(WMAG\)](#), Italian Alps (Sep 2022)
- Schools attended: [ERIS 2022](#), [ARIWS 2021](#), [ESCAPE 2021](#), [KITP 2021](#), [STScI 2022](#).